

# Research Paper 11 Summary

## Trends in Vegetation Fires in South and Southeast Asian Countries

**Problem Statement:** AI-Based Detection and Classification of Industrial Fires (PS26162)

**Original Paper:** Vadrevu et al. (2019), *Nature Scientific Reports*

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### 1 Basic Information

- **Paper Title:** Trends in Vegetation Fires in South and Southeast Asian Countries
- **Authors:** Krishna Prasad Vadrevu, Aditya Eaturu, Somnath Biswas, et al. (2019)
- **Journal:** *Scientific Reports* (Nature Publishing Group)
- **Focus Region:** India and South Asia across 15 years of satellite observations (2003–2017).

### 2 What is this Paper About? (In Simple Words)

This study provides the most complete calendar and regional map of fires across India. By analyzing 15 years of satellite records, the authors mapped the exact months and states where different types of fires occur.

- **The Indian Fire Calendar:**
  1. **Post-Rice (Kharif) Burning (October – November):** Massive concentration of stubble fires in Punjab, Haryana, and Western Uttar Pradesh.
  2. **Post-Wheat (Rabi) Burning (April – May):** Widespread crop fires across Central and Northern India.
  3. **Peak Forest Fires (February – May):** Dry-season forest fires in Central India (Madhya Pradesh, Chhattisgarh), the Western Ghats, and the Northeast.
  4. **Monsoon Season (June – September):** Vegetation and agricultural fires drop to near zero due to heavy rains.

### 3 Key Concept: The Seasonal Rhythm of False Positives

Natural and agricultural fires follow a strict seasonal schedule. In contrast, industrial facilities (refineries, steel mills, power plants) operate continuously all 12 months of the year, including during the monsoon.

*If a heat source continues emitting thermal signatures during July and August (monsoon months), it is virtually guaranteed to be an industrial facility, because wildland fires cannot survive heavy monsoon rainfall.*

## 4 How This Helps Our Project (PS26162)

### 4.1 1. Engineering Seasonal Features for the AI Model

We use the fire calendar from this paper to build cyclical temporal features for our machine learning classifier:

$$\text{Month}_{\sin} = \sin\left(\frac{2\pi \times \text{Month}}{12}\right), \quad \text{Month}_{\cos} = \cos\left(\frac{2\pi \times \text{Month}}{12}\right) \quad (1)$$

$$\text{Is\_Peak\_Crop\_Season} = \begin{cases} 1 & \text{if Month} \in \{\text{Oct, Nov, Apr, May}\} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

### 4.2 2. Monsoon Baseline Calibration (Clean Industrial Signals)

During the Indian monsoon season (June to September), natural fires disappear completely. Our system can use satellite observations during these 4 months as a “clean baseline” to detect and catalog all permanent industrial heat sources across India with zero false positives.

### 4.3 3. Setting Regional Prior Probabilities in GIS

The paper provides state-wise fire distribution profiles. Our GIS system can apply geographic priors:

- An alert in rural Punjab during November starts with a high prior probability of being agricultural burning.
- An alert in an industrial corridor in Gujarat starts with a high prior probability of being industrial.